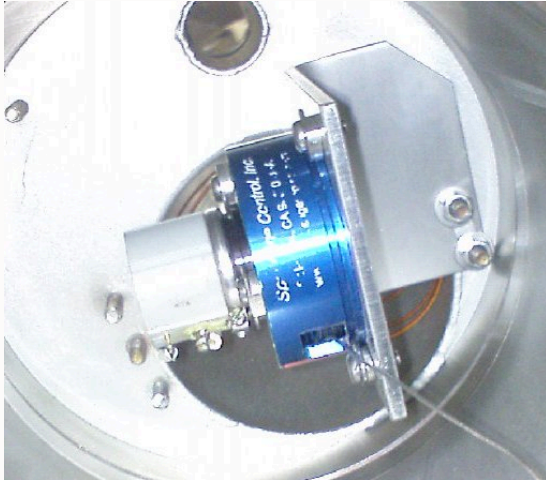


Møller Target Encoder Test

10 March 2010

There are results of the linear encoder (position transducers) readout test from epics variable HAMOL_LENC1 that have been performed on Feb 16, 2010. The picture of the linear encoder inside of the target frame:



Position	HAMOL_LENC1 value (V?)	Encoder Resistance (Ohm)
Parking (LL)	0.298	460
Target #1	1.731	2243
Target #2	2.011	2568
Target #3	2.281	2894
Target #4	2.554	3218
Extended (RL)	2.725	3470

Jog test (while the Target position switch is closed("1"))

Jog_Value	Encoder_Value
0	1.731
2.5	1.782

We have 14 inches between the limits and 2425 value in the encoder reading, so: $k = 25.4 \times 14 / 2425 = 0.1466$ mm/mV. Position in mm is $P = k \times \text{HAMOL_LENC1 (mm)}$. If Jog = 2.5 then distance while target on switch is $0.1466 \times (1782 - 1731) = 7.5$ mm.

Encoder values while in moving from Extended(LL) to Retracted(RL) Limits stored into file: *~moller/encoder_HAMOL_LENC1.txt*. The plot of this test is appended.

